

Reservoir stimulation's effect on depletion-induced seismicity

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Abstract. Depletion-induced seismicity can pose a problem in some fluid-producing subsurface reservoirs, in some cases requiring production rate control in order to limit the seismicity. This study investigates the use of reservoir stimulation to reduce the depletion-induced seismicity rate. Depletion-induced stress and pore pressure changes are evaluated in a shale caprock, sandstone reservoir, and shale underburden system which contains a horizontal well, all modelled in plane strain conditions. The seismicity rate is then predicted based on an existing seismicity model and is found to be dependent on the direction the well is drilled in with respect to the principal stresses. The case where the reservoir has first been stimulated is compared to the case where stimulation has not been performed (using the same production rates) for normal, reverse, and strike-slip faulting stress regimes. Seismicity is reduced in the case of reservoir stimulation for both reverse and strike-slip faulting stress regimes. The seismicity rate is only slightly reduced for the normal faulting stress regime. Stimulation also increases the distance that changes in pore pressure dominate over poroelastic stress changes in the reservoir. Further, it is found that the optimal orientation of a horizontal well, in terms of induced seismicity, is parallel to the minimum principal stress in normal faulting stress regimes and parallel to the maximum principal stress in reverse faulting stress regimes. The orientation of a horizontal well determines where the seismicity is located in a strike-slip faulting stress regime.

Keypoints:

- Well stimulation reduces depletion-induced seismicity.

- Horizontal wells have an optimum orientation in terms of induced seismicity.
- Stimulation increases distance over which changes in pore pressure dominate over those in stress.

Plain Language Summary

When fluid is removed from underground reservoir rocks during, for example, oil and gas production, it is observed that the stress changes caused in the reservoir by this fluid removal can cause earthquakes. This has been seen in places like Southwestern France, where large amounts of gas production have caused numerous instances of earthquakes large enough to be felt by the nearby public. In order to offer a potential partial solution to this problem, we investigated what it is specifically that causes these stresses. In essence, it is easier to remove fluid from certain kinds of rocks, and, the easier it is to remove fluid from a rock, the smaller the stress changes will be in the surrounding rock and the less likely it will be for earthquakes to occur. Therefore, we investigate whether implementing the use of hydraulic fracturing to more easily produce fluids from reservoir rocks would cause less stress changes and thereby less earthquakes. Our models show this to be the case; we suggest that producing fluid out of a fractured reservoir rock may actually result in fewer earthquakes than producing fluid out of an un-fracked reservoir rock.

Index Terms

1. 7299 General or miscellaneous
2. 5104 Fracture and flow

Keywords

1. Induced seismicity
2. Reservoir stimulation
3. Wellbore orientation

1. Introduction

During the last few years, induced seismicity due to fluid injection into subsurface reservoirs has become a major cause for concern despite its long-term record of observation [Ellsworth, 2013], with injection-induced earthquakes reaching moment magnitudes as large as 5.8 [Chen et al., 2017]. Whilst the primary focus has been on induced seismicity caused by changes in pore pressure [e.g., Layland-Bachmann et al., 2012; Zhang et al., 2013], poroelastic stresses are also thought to be a significant contributor in multiple cases of induced seismicity in both injection [see e.g. Barbour et al., 2016; Chen et al., 2017; Deng et al., 2016; Goebel et al., 2017] and production scenarios [Segall, 1985, 1989]. This has lead to poroelastic stresses being recognized as an important cause of induced seismicity and research has been performed on how to best account for these stresses such that induced seismicity is minimized [e.g., Chang & Segall, 2016a, b, 2017; Postma & Jansen, 2018; Segall & Lu, 2015].

Certainly, fluid extraction from subsurface reservoirs has long been known to be capable of inducing seismicity [Pratt & Johnson, 1926; Suckale, 2009, 2010]. Indeed, there are a number of specific examples of depletion-induced seismicity. The first reported case of depletion-induced seismicity occurred in Goose Creek, Texas, where oil and gas production lead to normal faulting on the flanks of the producing reservoir, causing a disturbance to nearby inhabitants [Pratt & Johnson, 1926; Yerkes & Castle, 1970, 1976]. The Wilmington oil field in the United States also exhibited reverse faulting at shallow depths above the reservoir during the primary production phase [Yerkes & Castle, 1976]. Gas production from the Gazli gas field is thought to have possibly induced reverse faulting underneath

the producing reservoir; the three largest induced events had a magnitude 7.0 in what was previously a seismically quiet region, injuring over one hundred people and resulting in one death [Simpson & Leith, 1985]. However, it is difficult to conclusively determine whether or not this event was related to the nearby gas production [Suckale, 2010]. The center of the Buena Vista Hills oil field showed signs of aseismic reverse faulting at the surface as a result of oil production, leading to significant pipeline and casing damage [Koch, 1933; Yerkes & Castle, 1970, 1976]. Hundreds of small to medium-sized earthquakes are thought to have been induced by very small poroelastic stress changes (no larger than 0.2 MPa) by production from the Lacq gas field in southwestern France [Segall et al., 1994], with most of the earthquakes occurring above the reservoir in this reverse faulting environment [Bardainne et al., 2008]. Gas production from a reservoir near Rocky Mountain House, Alberta induced reverse faulting ($M \leq 3.4$) below the producing reservoir [Wetmiller, 1986]. In fact, seismicity has been induced throughout the Western Canada Basin [Baranova et al., 1999]. Low magnitude reverse faulting ($M_w < 1.0$) occurred in Clinton County, Kentucky above and below a shallow producing oil reservoir [Rutledge et al., 1998]. These specific cases all can be explained by the model proposed by Segall [1989], which showed that poroelastic stresses induced by fluid extraction cause normal faulting on the flanks of reservoirs and reverse faulting above and below the reservoir. The poroelasticity model has also been used to show that the reverse faulting shocks which occurred underneath the Coalinga oilfield in California were probably not induced due to the relatively small stress changes fluid production was predicted to have induced near the earthquake hypocenters [Segall, 1985]. Additionally, it has been shown that normal faulting can also be induced within the reservoir if the stress path (i.e., the ratio of the

change in horizontal stress magnitudes to the change in pore pressure) is high enough [Chan & Zoback, 2002; Segall & Fitzgerald, 1998; Zoback & Zinke, 2002]. Examples of this type of seismicity include the Valhall and Ekofisk oilfields [Zoback & Zinke, 2002], where normal faulting has been seen within the reservoir, and the normal faulting within the Groningen gas field [Dempsey & Suckale, 2017]. It has also suggested that normal faulting occurring within the reservoir in the War-Wink gas field in West Texas may be explained by a high stress path [Zoback & Zinke, 2002]. Previously, Doser et al. [1991] found that the poroelasticity model presented by Segall [1989] was unable to explain the seismicity of the field. The specific case of depletion-induced seismicity related to a high stress path will not be further investigated here.

In the fully coupled theory of poroelasticity, pore pressure changes result in changes in total stress and changes in total stress result in pore pressure changes [see Biot, 1941; Rice & Cleary, 1976; Wang, 2000]. Specifically, it is the gradient of pore pressure that acts as an internal force in the equation of conservation of momentum and ultimately produces the deformation and poroelastic stress changes in the rock [Chang & Segall, 2017]. This implies that, if operators of gas, oil, geothermal, wastewater-disposal, or CO₂-sequestration wells were able to influence the gradient of pore pressure in the reservoir, they might also be able to influence the poroelastic stresses that are induced by their operations. Seeing how these stresses induce the seismicity as previously mentioned, it is probable that influencing the pore pressure gradient would also help them influence the seismicity rate. Hydraulic fracturing and acidization treatments are two forms of well and reservoir stimulation that could be used to improve the inflow performance of the

near-well region such that a smaller pore pressure gradient is required to produce the same amount of fluid [Dake, 1978; Economides & Nolte, 2000].

Therefore, this work shows how reservoir stimulation may reduce induced changes in total stress and thereby reduce the induced seismicity during given production scenarios, an extension of the idea that subsidence depends on diffusivity [Segall, 1985]. The analysis is performed for normal, reverse, and strike-slip faulting stress regimes with production from a horizontal well by assigning initial stresses based the notion of a critically-stressed crust [see Brudy et al., 1997; Townend & Zoback, 2000; Zoback & Townend, 2001; Zoback et al., 2002] and the idea that it is the frictional strength of pre-existing faults that limits differential stress; meaning that, for example, in a normal faulting environment with the vertical stress determined by the overburden, the minimum horizontal stress of a given stratum will be limited by the coefficient of friction of the faults that are likely to be present within that stratum [see Brudy et al., 1997; Zoback & Harjes, 1997; Zoback & Healy, 1992]. Therefore, the relatively low coefficients of friction seen in clay-rich rocks [Ewy et al., 2003], will result in the shales being able to support less differential stress. A sequentially coupled poroelastic flow simulator based on the continuity equation for mass balance [Lake, 1989], the conservation of momentum, and the linear theory of poroelasticity [Biot, 1941; Chang & Segall, 2016a; Rice & Cleary, 1976; Wang, 2000] is used to predict production-induced pressure and stress changes. Hydraulic fractures are modelled using relevant parameters from various hydraulic fracture treatments to produce an effective permeability near the well. The pressure and stress changes induced by reservoir pressure drawdown are then used as inputs for a seismicity model based on Dieterich [1994]. In this way, the pore pressure changes, stress changes, and seismicity

induced by production from the stimulated reservoir can be compared to the original, non-stimulated case. Specifically, the analysis of the effect of stimulation on induced pore pressure and stress changes that occur during production will provide insight into why stimulation is a production management tool with the potential to reduce induced seismicity in production reservoirs.

2. Methods

2.1. Coupled poroelastic flow model

Pore pressure and stress perturbations caused by fluid production are modelled with a poroelastic reservoir simulator which sequentially couples a finite volume flow model to a finite element linear elastic mechanical model. A 2D plane strain setup will be used. This is appropriate for the modelling of a horizontal well [Cheng, 1998]. Although the simulation is 2D, it will still be possible to calculate 3D stress changes [Cheng, 1998].

2.1.1. Flow model

Flow is governed by the balance equation for conservation of mass of a single phase, or the continuity equation [Lake, 1989], in combination with Darcy's Law,

$$\frac{\partial(\phi\rho)}{\partial t} - \nabla \cdot \left(\frac{k}{\mu} \rho (\nabla P - \nabla(\rho g z)) \right) = q. \quad (1)$$

Here, ρ represents the fluid density, which is compressible, μ the fluid dynamic viscosity, P the pore pressure, ϕ the reservoir porosity, k the permeability, g the acceleration due to gravity, z the depth, and q the mass source terms which are dictated by well injection and production mass rates.

Because of its non-linear nature, the equation will be solved numerically. To do this, Equation 1 is discretized in a residual finite volume in space and fully implicit finite difference in time framework [Aziz & Settari, 2002] in order to solve for an update to the primary variable, pressure. Note that the fluid properties at the interfaces (except in the gravity term) are upstreamed. All model boundaries have no flow, Neumann boundary conditions, except the top which has a constant pressure, Dirichlet boundary condition.

2.1.2. Mechanical model

The conservation of momentum is written as

$$\nabla \cdot \sigma' + \nabla (\alpha P) = -f, \quad (2)$$

neglecting inertial terms, where σ' is the effective stress, α is the Biot coefficient, P represents the pore pressure whose gradient acts as an internal force, and f represents the body forces. Compression and contraction will be taken as positive. This is then combined with the linear theory of poroelasticity [Biot, 1941; Chang & Segall, 2016a; Rice & Cleary, 1976; Wang, 2000],

$$S_{ij} - \alpha P \delta_{ij} = \frac{E}{(1 + \nu)} \epsilon_{ij} + \frac{E\nu}{(1 + \nu)(1 - 2\nu)} \epsilon_{kk} \delta_{ij}, \quad (3)$$

where S is the total stress, E the drained Young's Modulus, ν the drained Poisson's Ratio, ϵ the strain, and δ_{ij} the Kronecker delta. The equations are then discretized in a finite element framework with linear basis functions and quadrilateral elements situated such that the nodes are located at the corners of the finite volume cells from the flow model. The poroelastic properties E , ν , and K_s , the solid bulk modulus, are defined for the system, allowing for the calculation of α [Detournay & Cheng, 1993] via the calculation of a drained bulk modulus. All boundary conditions have a fixed perpendicular displacement, Dirichlet boundary condition except the surface, which is free.

2.1.3. Coupling

Here, the poroelastic coupling is performed using the definition of Lagrangian porosity from Coussy [2007],

$$\phi = -\epsilon_{kk} \alpha + \frac{P(\alpha - \phi_0)}{K_s} + \phi_0. \quad (4)$$

Here, ϕ is the current porosity, ϵ_{kk} is the volumetric strain, positive in contraction, and ϕ_0 is the reference porosity at a reference pore pressure and volumetric strain. The volumetric strain is a sum of the axial strains. The pore pressures found after the convergence of the flow model (Equation 1) are applied as internal forces in the mechanical model (Equations 2 and 3). The mechanical model then finds the new volumetric strain based on these pressures. This volumetric strain is then applied back in the flow model, changing the porosity. This means the fluid density (and therefore pressure) has to be adjusted such that mass is conserved. In this way the flow and mechanical model are iterated during each time step until convergence (determined by change in pressure) before moving on to the next time step.

2.2. Seismicity Model

The seismicity model is an adaptation based on an original formulation by Dieterich [1994] [Segall & Lu, 2015]. In this formulation, the Coulomb stress, τ , is defined as,

$$\tau = \tau_s - \mu_f (S_n - P), \quad (5)$$

where the shear stress on the fault plane is given by τ_s , the normal total stress on an optimally oriented fault in a given stress regime (negative in tension) by S_n , and the static coefficient of friction by μ_f . A variable R is then defined as the ratio of the current rate of seismicity to the background level and obeys the following relationship,

$$\frac{dR}{dt} = \frac{R}{t_a} \left(\frac{\dot{\tau}}{\dot{\tau}_0} - R \right). \quad (6)$$

Here, $\dot{\tau}$ represents the stressing rate, $\dot{\tau}_0$ is the background stressing rate, and t_a is the characteristic decay time which is taken as 50 years, matching other authors [Segall &

Lu, 2015]. This seismicity model uses the stress and pore pressure updates coming from the poroelastic simulator to find a new stressing rate at the end of each time step. This is then used to iteratively solve for the new seismicity rate in an implicit finite difference manner by setting Equation 6 into a residual formulation,

$$r^\omega = \frac{R^\omega - R^n}{\Delta t} - \frac{R^\omega}{t_a} \left(\frac{\dot{\tau}^n}{\dot{\tau}_0} - R^\omega \right) = -\frac{\partial r}{\partial R} |^\omega \Delta R^{\omega+1}. \quad (7)$$

Here, r represents the value of the residual, n represents the previous time step, ω represents the current iteration, Δt is the time step size, and the update to the primary variable, R , is given by ΔR . Note that $\dot{\tau}^n$ is the average stressing rate over the time step. The minimum value for R is assigned to be 0.01. The seismicity rates produced by this model are used to estimate how seismicity responds to different production scenarios. Certainly, it can be deduced from Equation 5 that increases in pore pressure, increases in the maximum principal stress, and decreases in the least principal stress will all increase the Coulomb stress. Further, from Equation 6 it is apparent that increases in the Coulomb stress will cause a higher seismicity rate. In this way it is possible to have an intuition for the effect a given set of pressure and stress changes will have on seismicity rate. Note that all lithologies present in the model will use this seismicity model; however, broadly speaking, each lithology's unique coefficient of friction and initial stress profile will govern how significant a change in pore pressure or stress will be in terms of induced seismicity.

3. Problem Setup

Production through a horizontal well in a plane strain case in a 378 meter thick, water-saturated sandstone reservoir with a shale cap rock and underlying shale layer is considered; see Figure 1. The thickness of the reservoir was chosen to match the range of the

effective reservoir thickness seen at the Lacq gas field [Segall et al., 1994]. The producing reservoir being enclosed in the vertical direction by two shales is justified as this is similar to real examples of petroleum reservoirs (e.g., Coalinga [Segall, 1985]). In fact, in many practical applications the rocks surrounding the producing reservoir in oil and gas applications have very low permeabilities [Segall, 1992]. Production in this case occurs through a horizontal well as this type of well has been frequently used since the 1990's [Nurmi et al., 1995; Zhang & Dusseault, 1994]. A 2D plane strain setup is appropriate to model these kinds of conditions and allows for the calculation of stress changes in all three principal directions [Cheng, 1998]. Example normal, reverse, and strike-slip faulting stress regimes are considered. In each case, the well is drilled parallel to the minimum principal stress based on an argumentation regarding fracture development [Hubbert & Willis, 1957], except in the reverse faulting case where the well is drilled parallel to the intermediate principal stress; see Table 1. The initial pore pressure and vertical stress are found through standard hydrostatic and lithostatic gradients respectively. Depending on the stress regime, either the initial maximum or initial minimum principal stress is then found using the vertical stress, pore pressure, and the assumption that the crust is critically stressed [see Brudy et al., 1997; Townend & Zoback, 2000; Zoback & Townend, 2001; Zoback et al., 2002] with the frictional strength of pre-existing faults limiting the differential stress supported by the crust [see Brudy et al., 1997; Zoback & Harjes, 1997; Zoback & Healy, 1992]. In the strike-slip faulting stress regime, it is first assumed that the minimum principal total stress is equivalent to 0.8 times the initial vertical total stress [Zoback, 2007]. Production from the reservoir, whether it is stimulated or not, will occur over a period of 10 years from a 750-meter section of a horizontal well, a typical lateral

section length of a horizontal well [Joshi, 1991], and with exactly the same production rate in all cases (beginning at $2e5 \frac{kg}{year \cdot m}$, or approximately $2500 \frac{bbl}{day}$ for this well). The production rate will remain constant during a plateau period of three years before declining at a rate of 6% per year, not dissimilar to the production decline behaviour that is seen in some oil fields [Höök et al., 2009], Figure 2a. All test cases begin with an undisturbed pressure and stress field at the reservoir scale. The seismicity that might be incurred due to the process of hydraulic fracturing itself is not included in this analysis because hydraulic fracturing is not expected to induce seismicity at lateral distances larger than about 400-1000 meters [Rubinstein et al., 2018; Westwood et al., 2017], and the seismicity in the reservoir being evaluated here is occurring at least 1500 meters from the well. Also, although there are some exceptions, the seismicity occurring due to hydraulic fracturing only lasts a few days after the stimulation treatment [Lei et al., 2017; Rubinstein et al., 2018], whereas the analysis here is occurring over ten years. A full list of parameters used to perform the simulations can be found in Table 2. Note that, although different parameters will change the quantitative results, the qualitative effects that stimulation has on depletion-induced seismicity generally remain the same.

3.1. Modelling Reservoir Stimulation

In order to model a stimulated permeability field resulting from a hydraulic fracture treatment, parameters for a typical treatment were used to predict the size and effect of the resulting fracture. The height of the hydraulic fracture was estimated based on Flewelling et al. [2013], who used fracture mechanics models to estimate the height of a simple planar fracture such that,

$$H = \left[\frac{3\Omega V E}{\pi a_r P_n (1 - \nu^2)} \right]^{\frac{1}{3}}. \quad (8)$$

Here, H is the fracture height, V the volume of the fracture, a_r the fracture aspect ratio, which will be taken as 1.36, a reasonable value according to Smith et al. [2001], P_n the net pressure which is the difference between the fluid pressure in the fracture and the least principal stress, and Ω is a shape parameter which is given by $\Omega \cong \sqrt{1 + 1.464a_r^{1.65}}$ [Anderson, 2005]. Typically, the value of $\frac{E}{P_n}$ is around 10^3 at the borehole [e.g., Fisher et al., 2002; Warpinski et al., 1990]. However, recognizing that this value may vary along the fracture, Flewelling et al. [2013] use a range between 6,000 and 30,000. Here, we will take a value of 22,000. The volume of the fracture can be calculated using the volume of the fluid injected and the leakoff, taken here as 20%, a reasonable value for conventional hydraulic fracturing [Shen et al., 2011]. The volume of fluid injected per stage will be taken as $2,338 \text{ m}^3$, which, based on a stage length of 100 m across a 750 m well, falls within the range of amount of fluid used to hydraulically fracture an average well in the Eagle Ford Shale [Nicot & Scanlon, 2012]. Based on these values and consistent with field observations reported by Flewelling et al. [2013], the fracture height is approximately 385 m, assuming a sufficiently thick homogeneous reservoir rock. Based on the assumed aspect ratio, the fracture width is 525 m. Note that it is assumed that the fracture will remain in the sandstone reservoir and not propagate into the overlying and underlying shale layers. This is because, due to their lower coefficients of friction [Ewy et al., 2003], shales are able to support less differential stress. As the vertical stress is determined by the overburden, this means that the minimum principal stress is predicted to be higher in shales than in sandstones in normal faulting environments [Zoback, 2007] based on

the notion of a critically stressed crust, discussed previously. A hydraulic fracture needs to overcome the minimum principal stress in order to propagate, so it can be assumed that the higher minimum principal stress in the shale will prevent fracture propagation [Zoback, 2007]. This is supported by experiments done by Warpinski et al. [1982], who showed that differences in the minimum principal stress are the primary control on fracture propagation through multiple layers. It is also assumed that the hydraulic fracture in the strike-slip faulting stress regime will be contained within the sandstone reservoir. As will be discussed below, the hydraulic fracture in the reverse faulting stress regime does not propagate vertically.

Stimulation treatments performed in the Rotliegendes sandstone of Central Europe in Enhanced Geothermal Systems have inferred apertures up to 10 *mm* [Zimmermann et al., 2015]. Although these stimulation treatments are being performed at significantly greater depths than would be typical of hydrocarbon production, they do give an indication of a typical aperture size and indicate that a more conservative aperture of 2 *mm* is reasonable. Next, in order to calculate an effective permeability of the reservoir blocks containing the fracture, the effective permeability of fractured media, given as [Roman et al., 2012],

$$k = \frac{a^3}{12L_0}, \quad (9)$$

is used, even though the smooth plate assumption inherent in this model is thought to overestimate the fracture permeability [Kluge et al., 2017]. Here, a is the fracture aperture and L_0 is the length of the porous media sample in the direction perpendicular to the fracture plane. Assuming that only one fracture forms per hydraulic fracturing stage, this length is then equivalent to the stage length, which is 100 *m* in this case. This results

in an effective permeability of approximately $6.67 \times 10^{-12} \text{ m}^2$ for all cells containing the fracture (Figure 2b and d). This is then the permeability for both the x- and z-directions in the model.

Note that in a reverse faulting case, the least principal stress is oriented vertically, so the fracture is expected to grow normal to the vertical direction. Although the fracture height will be the same, this height will actually be oriented in the horizontal direction. Further, the L_0 used in Equation 9 will now simply be the height of the reservoir grid block. The fracture will be modelled to not affect the vertical permeability and will be contained, in the z-direction, within one block, modelled with an effective permeability of $1.23 \times 10^{-11} \text{ m}^2$, calculated from Equation 9, Figure 2c.

Although this is a simple way to model a stimulation treatment, it is sufficient to investigate the effect of a general stimulation treatment on the induced seismicity seen during subsequent fluid production.

4. Results & Interpretation

4.1. Normal Faulting Stress Regime ($S_z > S_x > S_y$)

A hypothetical well has been drilled parallel to the minimum principal stress in a normal faulting stress regime, which in this case is in the y direction. As this is a normal faulting stress regime, focus will be placed on the model's results at the flanks of the reservoir, between 1.5 and 4 kilometers from the well. Beyond 6 kilometers from the well, pore pressure and stress changes become constant due to the cumulation of the applied forces resulting from the pore pressure gradients nearer the well, indicating that the model is sufficiently wide to avoid boundary effects influencing the results. The flanks of the reservoir and surrounding rock are the most likely place for induced seismicity to occur

in this stress regime [Segall, 1989]. Indeed the reason for this can be seen in Figure 3c, where the negative (i.e. less compressive) total stress changes and pore pressure changes for the system are shown for the case where no stimulation has been performed. All stress magnitudes are taken at the same depth as the well. Note that although the stress changes are negative, the resultant stresses are all still compressive (positive) overall. This is because the largest stress changes are less than 10 MPa, which is significantly smaller than the initial stresses at that depth. At locations on the flanks of the reservoir which are within the reservoir but far away from the well (beginning around 2km from the well), tensile changes are induced in S_y , which is the least principal stress. The maximum principal stress, S_z , experiences the least amount of tensile change. The decrease of the least principal stress destabilizes any potential optimally oriented faults in the reservoir, while the tension induced in the maximum principal stress is relatively insignificant due to the small magnitude of the stress change. There is also, however, a decrease in pore pressure which helps to stabilize the reservoir. Overall, at large distances, the stress changes dominate and the reservoir is expected to destabilize as can be seen in Figure 3a. Poroelastic stress changes dominating over pore pressure changes and inducing seismicity at large distances also agrees with the findings of other authors [e.g., Chang & Segall, 2016a, b; Goebel et al., 2017; Segall & Lu, 2015].

The Coulomb stress changes that occur in the stimulated and non-stimulated cases can be seen in Figure 3d. Stimulation has the effect of reducing the pore pressure (after the first 185 m from the well - closer to the well pore pressure is actually higher in the stimulated case) and induces lower (more tensile) stresses than in the non-stimulated case. A lower pore pressure and a more tensile maximum principal stress (represented by the changes

to S_z) are expected to reduce the seismicity; however, the changes to S_z are relatively small and the least principal stress, S_y , also becomes more tensile. Ultimately, as can be seen in Figure 3b, the seismicity is reduced due to large reductions in pore pressure far from the well, but the changes to S_y serve to mitigate this reduction in seismicity and the seismicity rate is only moderately reduced.

4.2. Reverse Faulting Stress Regime ($S_x > S_y > S_z$)

In a reverse faulting stress regime, with the well drilled parallel to the intermediate principal stress, the low pore pressures and decrease in S_x (the maximum principal stress) in the reservoir imply that the Coulomb stress and the seismicity in the reservoir are reduced. As can be seen in Figure 4c, however, above and below the reservoir S_x is increased. Poroelastic effects also mean that the pore pressure is increased at these locations. These two effects combine with a decrease in S_z , the least principal total stress, to cause Coulomb stress increases above and below the reservoir, resulting in an increase in seismicity, as seen in Figure 4a and predicted by [Segall, 1989].

As can be seen in Figure 4d, stimulation has the effect of reducing the Coulomb stress changes occurring above and below the reservoir. The higher diffusivity near the well reduces the tension induced in S_z as well as the compression induced in S_x above and below the reservoir. Further, it results in a reduction in the induced poroelastic pore pressure increase at these locations. All of these changes reduce the Coulomb stress above and below the reservoir and therefore also the seismicity rate, which can be seen in the comparison made in Figure 4b for selected locations above and below the reservoir. At each location the seismicity is reduced for the case that the reservoir was stimulated.

4.3. Strike-Slip Faulting Stress Regime ($S_x > S_z > S_y$)

In the case that a well is drilled in a strike-slip faulting stress regime (with S_y being the least principal stress), the production-related reduction in S_x occurring in the reservoir along with the production-related reduction in pore pressure, again mean that there is a reduction in seismicity occurring within the reservoir during production. Above and below the reservoir, however, the horizontal stresses increase, especially S_x , and, due to poroelastic effects, pore pressure also increases at these locations. The pore pressure and S_x increase lead to higher Coulomb stresses, which leads to a seismicity rate increase as seen in Figure 5a. The seismicity is less than in the reverse faulting stress regime case because the minimum stress, S_y , actually increases in this stress regime whereas in the reverse faulting stress regime the minimum stress, S_z , decreases due to production.

As seen in Figure 5b, in the case the reservoir is stimulated before production begins, subsequent depletion-induced seismicity is reduced. Two locations are evaluated in Figure 5b, just above and just below the reservoir. In both locations, the seismicity is reduced by a significant amount. This occurs because less Coulomb stress changes are induced in the case that the reservoir has been stimulated, as seen in Figure 5d. This is especially due to the reduction in the amount of compression in S_x that is occurring, which can be seen in Figure 5d, and also due to the lessening of the pore pressure increase, although this effect is of lesser importance than the change to S_x . The least principal stress, S_y , also experiences less compression due to the stimulation treatment (which is an undesired effect in terms of induced seismicity); however, the changes to S_x are again more significant.

4.4. Optimal Well Orientations

Based on the differences in horizontal stress changes seen in Figures 3c, 4c, and 5c, it is intuitive that the seismicity levels are likely to depend on the direction that a horizontal well is drilled in. In this section, the results for the non-stimulated case will be rerun with the wells drilled perpendicular to their original orientations. Because this would change how a hydraulic fracture would propagate, only the non-stimulated cases will be considered.

4.4.1. Normal Faulting Stress Regime

If the well were to be drilled in the direction of the intermediate principal stress instead of the least principal stress in a normal faulting stress regime, S_x would represent the changes to the least principal stress in Figure 3c. Seismicity on the flanks of the reservoir would be higher due to the even higher Coulomb stress (Figure 6a), because changes to S_x are larger than changes to S_y in the areas that experience seismicity. This agrees with the results of Altmann et al. [2014], who showed that radial stress changes are generally larger than tangential ones in production scenarios. Further, this implies that, if the goal is to minimize induced seismicity, in a normal faulting stress regime, horizontal wells should be drilled parallel to the least principal stress to minimize the induced Coulomb stress resulting from fluid production on optimally-oriented faults at the flanks of the reservoir. This is confirmed in Figure 6b.

4.4.2. Reverse Faulting Stress Regime

The seismicity in a reverse faulting stress regime depends on the direction that the horizontal well is drilled in. If the well were to be drilled parallel to the maximum principal stress, seismicity would likely be reduced above and below the reservoir. The reason for

this can be seen in Figure 4c where compressive changes to S_x are larger than those to S_y .

This means that drilling the well such that it is parallel to the maximum principal stress reduces the compression induced in the maximum principal stress, resulting in a smaller increase in the Coulomb stress, as seen in Figure 7a. Therefore, in terms of induced seismicity, it is beneficial to drill horizontal wells parallel to the maximum principal stress in a reverse faulting stress regime, as shown in Figure 7b. Regardless of well orientation in this stress regime, seismicity is not expected in the reservoir because in these locations more tension is induced in S_x and S_y than in the least principal stress, S_z , and pore pressure is reduced.

4.4.3. Strike-Slip Faulting Stress Regime

The results for strike-slip faulting stress regimes also depend on the direction that the horizontal well is drilled in. If the well were to be drilled parallel to the maximum principal stress as opposed to the least principal stress, the seismicity would be focused on the flanks of the reservoir, as shown in Figure 8. The reason for this can be seen in Figure 3c (which shows stress changes in the reservoir as opposed to above and below it as in Figure 5c) where tensile changes to S_x (in this case the least principal stress) are larger than the tensile changes to S_y (in this case the maximum principal stress) at the flanks of the reservoir, meaning that the Coulomb stress will increase at these locations. Seismicity would not occur above and below the reservoir, based on the stress changes occurring above and below the reservoir in Figure 5c, because in these locations the compressive changes to S_x are larger than those to S_y .

5. Discussion and Conclusions

In a normal faulting stress regime, fluid production causes a reduction in the minimum principal stress and pore pressure at the flanks of the reservoir and a slight reduction in the vertical stress. The reduction in the minimum principal stress is what leads to seismicity at these locations. Drilling the horizontal well in a different orientation than parallel to the least principal stress, which is horizontal, is predicted to increase the induced seismicity. Indeed, it has been previously shown that an anisotropic stress field can have an effect on how seismicity develops [Schoenball et al., 2010]. Wellbores in horizontal wells drilled in the direction of the least principal stress are also more stable than those drilled in the direction of the secondary principal stress [Zoback, 2007]. In the case that the reservoir is stimulated before production, the pore pressure distribution begins at a higher value near the well. It also has a much lower pore pressure gradient near the well, resulting in less induced tension at these locations. However, after a certain distance (in this simulation, 185 *m* away from the well), the pore pressure is actually lower in the stimulated case. This results in increased mechanical stability in these areas. That said, pore pressure also has a steeper gradient far from the well which results in more induced tension in S_y , an effect which would normally be destabilizing for optimally oriented faults. However, as the stimulation's effect on production-induced pore pressure changes is greater than its effect on production-induced changes in S_y , the stimulation has the effect of making these areas more stable. Although a small effect, the tension induced in S_z at the flank of the reservoir is also increased due to stimulation, somewhat stabilizing the area. Overall the stimulation has the effect of slightly reducing the induced seismicity caused by fluid production.

Further, because the effect of the fracture's presence is larger on the pore pressure distribution than on the stress changes over most of the reservoir, stimulation effectively increases the distance that pore pressure changes dominate over stress changes in terms of induced seismicity. In this numerical experiment, when compared to the non-stimulated case, pore pressure changes were larger than changes to S_x some 34 m farther away from the well in the case that the reservoir was stimulated. In this case this change is small, but this value is dependent on the simulation parameters. Qualitatively, the increase in distance that pore pressure changes are larger than stress changes can be predicted using the analytical solutions for pore pressure and stress due to fluid injection/production in an isotropic body of Rudnicki [1986], where the similarity variable, ξ , is

$$\xi = \frac{r_d}{\sqrt{ct}}. \quad (10)$$

Here, r_d is the radial distance from the point source, t is time, and c is the hydraulic diffusivity, given by,

$$c = \frac{k (\lambda_u - \lambda) (\lambda + 2G)}{\mu \alpha^2 (\lambda_u + 2G)}. \quad (11)$$

Here, μ is the fluid dynamic viscosity, k is the permeability of the isotropic medium, λ_u and λ are the undrained and drained Lamé parameters, respectively, and G is the shear modulus. Then, from Rudnicki [1986], the ratio of the pore pressure change to the stress change in the direction of a principal stress, in this case taking S_x as an example, is given by,

$$\frac{P}{S_x} = \frac{\mu c \alpha (\lambda_u + 2G) \operatorname{erfc} \left(\frac{1}{2} \xi \right)}{k (\lambda_u - \lambda) G \left[2 \operatorname{erfc} \left(\frac{1}{2} \xi \right) + 4 g \xi^{-2} \right]}, \quad (12)$$

where g is

$$g = \operatorname{erf}\left(\frac{1}{2}\xi\right) - \frac{\xi}{\sqrt{\pi}}e^{-\frac{1}{4}\xi^2}. \quad (13)$$

Based on Equation 12, it is clear that the ratio of pore pressure changes to changes in S_x is dependent on ξ . Therefore, based on Equations 10 and 11, the location where these two sets of changes are equal (i.e. where the ratio is equal to 1) will change as permeability changes. Specifically, the location where the pore pressure change is equivalent to a change in a principal stress increases with the square root of permeability, as illustrated in Figure 9. It should be noted, however, that the results of Rudnicki [1986] are for a point source of fluid mass in an isotropic and homogeneous medium, meaning they are not valid for predicting the increase in distance that pore pressure changes are larger than stress changes due to a hydraulic stimulation. That said, they do imply generally that increasing the permeability of a reservoir will increase this distance.

In a reverse faulting stress regime, the seismicity is primarily seen above and below the reservoir during production. This is because, in these locations, the maximum principal stress, S_x , increases and the least principal stress, S_z , decreases. Additionally, pore pressure increases at these locations due to poroelastic effects. All of these changes have an adverse influence on the Coulomb stress and therefore the seismicity rate. In the case where the reservoir has been stimulated, the smaller resultant pore pressure gradients result in all three of these changes being lessened. This then reduces the amount of seismicity expected. The optimal orientation of horizontal wells drilled in this stress regime, in terms of induced seismicity, is found to be parallel to the maximum principal stress,

which is horizontal. This is also the optimal direction of a horizontal well in this stress regime for borehole stability [Zoback, 2007].

Finally, in a strike-slip faulting stress regime with a horizontal well drilled parallel to the least principal stress, seismicity is seen above and below the reservoir during production but to a lesser extent than in the reverse faulting case. Here, the maximum principal stress, S_x , is increased and the pore pressure is also increased due to poroelastic effects, both decreasing stability. However, the least principal stress, S_y , is also increased which acts to stabilize these areas. Seismicity still occurs, however, as the changes in S_x are much larger. Stimulating the reservoir acts to reduce all of these changes in pressure and stress, and ultimately reduces the amount of induced seismicity. Drilling the well parallel to the maximum principal stress changes the location of the induced seismicity to the flanks of the reservoir. Drilling a horizontal well in the direction of the maximum principal stress in a strike-slip faulting regime is favourable in terms of well bore stability [Zoback, 2007].

The reduction of seismicity in fluid-producing reservoirs with stimulation, while demonstrated for a permeability field resulting from a hydraulic fracturing treatment, could be expected to be seen as a result of acidization treatments as well. This is because these type of treatments all act to increase the permeability (and therefore transmissibility) of the region around the well, allowing the same amount of fluid to be produced with a smaller drawdown. This reduces the changes in poroelastic stresses and pressure occurring above and below the reservoir. Stimulation also allows the reduction in pore pressure to propagate farther more quickly within the reservoir, stabilizing areas far from the well but still within the reservoir. While it is true that this results in higher pore pressure gradients far

from the well (inducing more poroelastic stress changes which act to destabilize faults in a normal faulting stress regime), it also increases the distance that pore pressure changes dominate over poroelastic stress changes.

Conversely, depletion-induced compaction is likely to reduce the permeability of producing reservoirs over long periods of time [Schutjens et al., 2004]. This will presumably result in higher pore pressure gradients required to maintain the same production rate, which suggests that an increase in Coulomb stress and therefore seismicity rate will result, but this should be investigated. Further, the results imply that, generally speaking, in the case that equivalent flow rates are used to produce two reservoirs which are identical in all aspects bar their permeabilities, the reservoir with the lowest permeability should be expected to experience more seismicity, agreeing with the results related to subsidence in Segall [1985].

These results depend strongly on the stress regime. The stimulation treatment affects each of the principal stresses uniquely, and, therefore, the stimulation's effect on the induced seismicity will also differ in each of the different possible stress regimes depending on the orientation of the well. Indeed, stimulation is most effective at reducing seismicity in reverse and strike-slip faulting stress regimes and has only a moderate effect in a normal faulting stress regime. These results are also dependent on the parameters used during the simulation. Although, qualitatively speaking, the effects of the stimulation will be the same, certain cases may experience larger seismicity rate reductions than others. Further, it should be noted that the seismicity model used does not provide insight into the size of the earthquakes induced.

The seismicity modelled here was induced with small Coulomb stress changes (on the order of 0.1 MPa or even less in the normal faulting stress regime case), but this is in agreement with what is seen in real life production scenarios. For example, 0.2 MPa stress changes were seen to be enough to induce felt seismicity in the Lacq gas field, France [Segall et al., 1994], and 0.02 MPa stress changes induced microseismicity in the Seventy-Six oil field, Clinton County [Rutledge et al., 1998]. In some cases, even a stress change of 0.01 MPa may be sufficient to induce seismicity [Hardebeck et al., 1998; Reasenbergs & Simpson, 1992]. Indeed, it is generally thought that a stress change of 0.1 MPa is enough to induce an earthquake under the correct circumstances [Stein, 1999].

Cases where small stress changes result in induced seismicity support the assumption of a critically-stressed crust used here. It is thought that most of the crust is critically stressed, even intraplate areas which do not often experience seismicity [Townend & Zoback, 2000; Zoback, 2007]. However, intraplate earthquakes' infrequent occurrence may even make them more hazardous as these areas may not be prepared for seismicity. Intraplate areas are seismically relatively quiet only because of low strain rates [Zoback, 2007; Zoback et al., 2002], and many intraplate areas do experience significant seismicity over long periods of time, including events with a large ($M_w \geq 7$) magnitude [Talwani, 2014; Zoback & Gorelick, 2012]. In fact, the deadliest earthquake in human history was an intraplate earthquake [Talwani, 2014]. Due to the low stressing rates seen in intraplate areas, even small stress changes may significantly increase the chance of seismicity [Parsons, 2005]. This helps explain why seismicity can be induced by fluid production in areas that were previously not associated with seismicity, such as in the Gazli gas field [Simpson & Leith, 1985], Goose Creek [Yerkes & Castle, 1976], the Lacq gas field [Segall et al., 1994],

and the Western Canada Sedimentary Basin [Baranova et al., 1999], including Rocky Mountain House [Wetmiller, 1986]. Further, most intraplate areas are characterised by either reverse or strike-slip faulting stress regimes [Zoback et al., 1989] and many (approximately 40%) of the world's largest oil fields are found in tectonic settings which are characterised by reverse and strike-slip faulting [Mann et al., 2001]. This makes the results presented here especially relevant, as stimulation was shown to have the largest induced seismicity-reducing effect in these two stress regimes.

It is important to recognize that these results do not necessarily mean that reservoir stimulation will result in lower seismicity rates if the reservoir is then produced at a higher rate than it would have been without the stimulation treatment, as all simulations used the same producing mass flow rate. In effect, the simulations ran are equivalent to stimulating a reservoir for the sole purpose of reducing the seismicity and not in order to increase the production rate. Additionally, it is important to note that the act of hydraulically fracturing itself has been known to induce seismicity [e.g. Atkinson et al., 2015; Clarke et al., 2014; Deng et al., 2016; Wang et al., 2017], although the seismicity associated with hydraulic fracturing is usually seen to return to a low level shortly (on the order of days - much shorter than the typical production life of a conventional well) after the hydraulic fracturing treatment [Lei et al., 2017; Rubinstein et al., 2018]. Further, the disposal of waste water either used to perform hydraulic fracturing treatments or co-produced with the hydrocarbons after stimulation treatments has also been seen to cause induced seismicity [e.g. Chen et al., 2017; Horton, 2012; Keranen et al., 2013]. These are serious concerns that should be carefully considered before proceeding with a hydraulic fracturing treatment. Instead, this paper qualitatively highlights the positive impact a

stimulated permeability field could have on depletion-induced seismicity occurring in and around the production reservoir.

These results could have implications for production wells in reservoirs experiencing depletion-induced seismicity. However, for large oil fields, multi-well simulations should be performed as these fields are typically produced with many wells as opposed to the one well example shown here. Finally, the manipulation of the pore pressure gradient to influence the induced poroelastic stresses and thereby reduce the seismicity is also something that might be achievable in other ways, such as through manipulating the fluid viscosity in injection scenarios.

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Table 1. *Principal stress orientations.* In each case the well is drilled in the y-direction;

however, the orientations of the principal stresses change depending on the stress regime.

Regime	S_1	S_2	S_3
Normal Faulting	S_z	S_x	S_y
Reverse Faulting	S_x	S_y	S_z
Strike-Slip Faulting	S_x	S_z	S_y

Table 2. *Model parameters*

Name	Variable	Value	Units
Model Width	L_x	24325	m
Cap rock height	$L_{z,c}$	1400	m
Reservoir height	$L_{z,r}$	378	m
Underlying shale height	$L_{z,us}$	2596	m
Grid cells, elements x	N_x	139	-
Grid cells, elements z	N_z	81	-
Production time	t_{prod}	10	years
Plateau flow rate	q_{flow}	2e5	$\frac{\text{kg}}{\text{year} \cdot \text{m}}$
Fluid reference density	ρ_0	1000	$\frac{\text{kg}}{\text{m}^3}$
Fluid compressibility [†]	c_f	5e-10	$\frac{1}{\text{Pa}}$
Fluid dynamic viscosity	μ	0.001	Pa · sec
Permeability shale [†]	k	1e-19	m ²
Permeability sandstone ^β	k	1e-15	m ²
Drained Young's Modulus shale ^χ	E	5e9	Pa
Drained Young's Modulus sandstone ^β	E	18e9	Pa
Drained Poisson's Ratio shale ^χ	ν	0.25	-
Drained Poisson's Ratio sandstone	ν	0.25	-
Grain bulk modulus shale ^Θ	K_s	6e9	Pa
Grain bulk modulus sandstone ^β	K_s	36e9	Pa
Initial porosity shale [†]	ϕ_0	0.23	-
Initial porosity sandstone ^β	ϕ_0	0.25	-
Coefficient of friction shale [†]	μ_f	0.35	-
Coefficient of friction sandstone ^Θ	μ_f	0.6	-
Background stressing rate ^Θ	$\dot{\tau}_0$	0.001	$\frac{\text{MPa}}{\text{year}}$
Characteristic decay time ^Θ	t_a	50	years

[†] Compressibility of water based on Osif [1988][†] Taken from Opalinus Clay (OPA) Shale [Orellana et al., 2018]^Θ Typical value [Segall & Lu, 2015]^χ Typical OPA Shale values, neglecting anisotropy [Favero et al., 2018]^β In range of typical sandstone values [Detournay & Cheng, 1993]^Θ Typical for clay minerals [Vanorio et al., 2003]

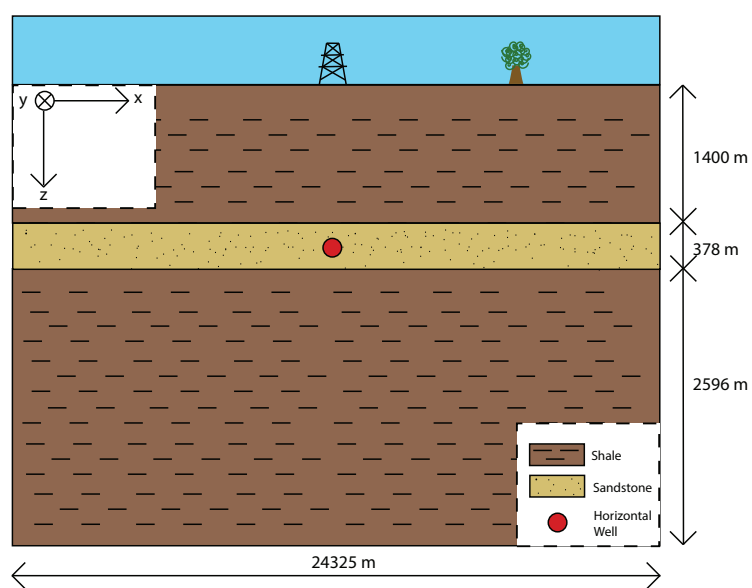


Figure 1. Problem overview, vertically exaggerated

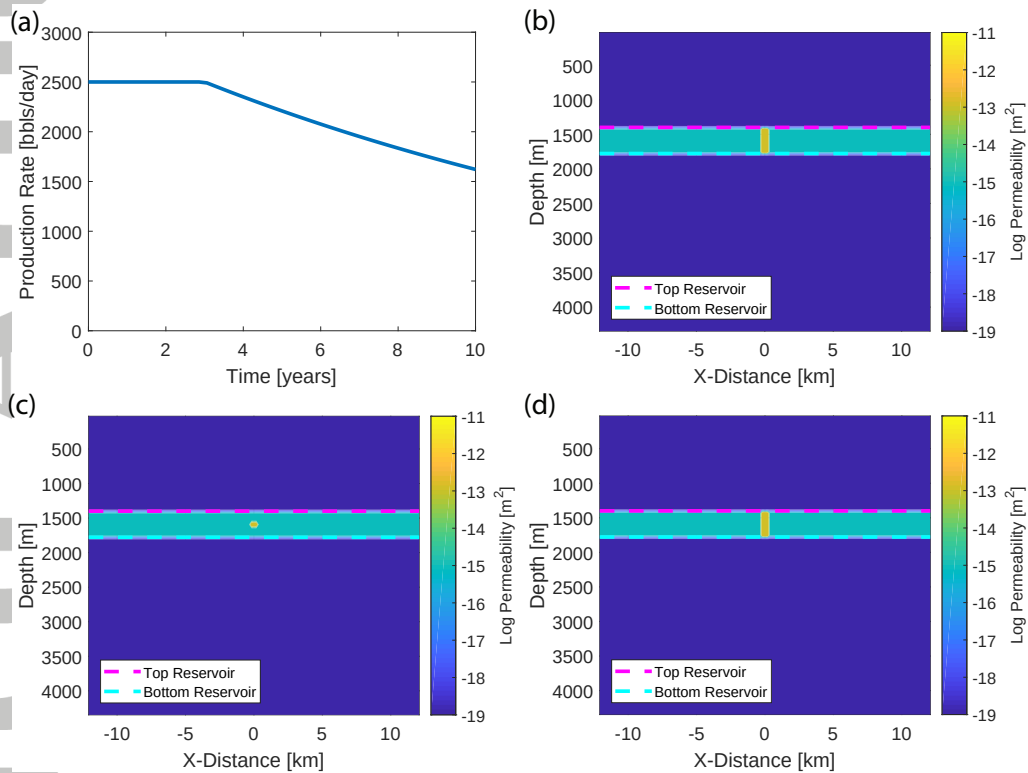


Figure 2. (a) The production rate throughout the simulation. The stimulated permeability fields for the (b) normal, (c) reverse, and (d) strike-slip faulting stress regimes. The simulated hydraulic fracture always propagates in a plane perpendicular to the minimum principal stress. The fracture is therefore normal to the y-direction in the normal and strike-slip faulting stress regime cases and normal to the z-direction in the reverse faulting case.

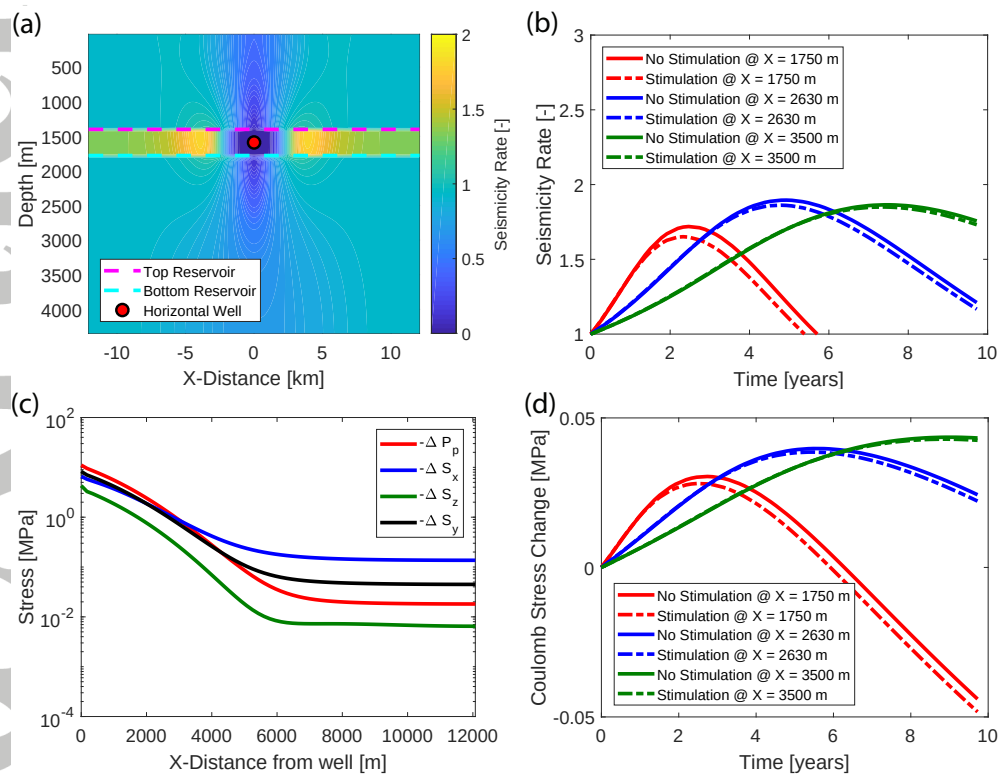


Figure 3. Results at the end of simulation ($t = 10$ years); (a) The seismicity rate relative to the background seismicity rate in a normal faulting stress regime without stimulation. Seismicity is generally concentrated at the flanks of the reservoir as predicted by Segall [1989]. (b) The seismicity rate at various locations at the same depth as the well with and without stimulation. Note that based on (a) these are the most seismically active areas in the simulation. (c) The reductions in pore pressure and the decreases in total stress due to production from the well in the case where no stimulation has occurred. Data are reported only for the locations at the same depth as the well. (d) The Coulomb stress changes for the stimulated and non-stimulated cases at various locations at the same depth as the well.

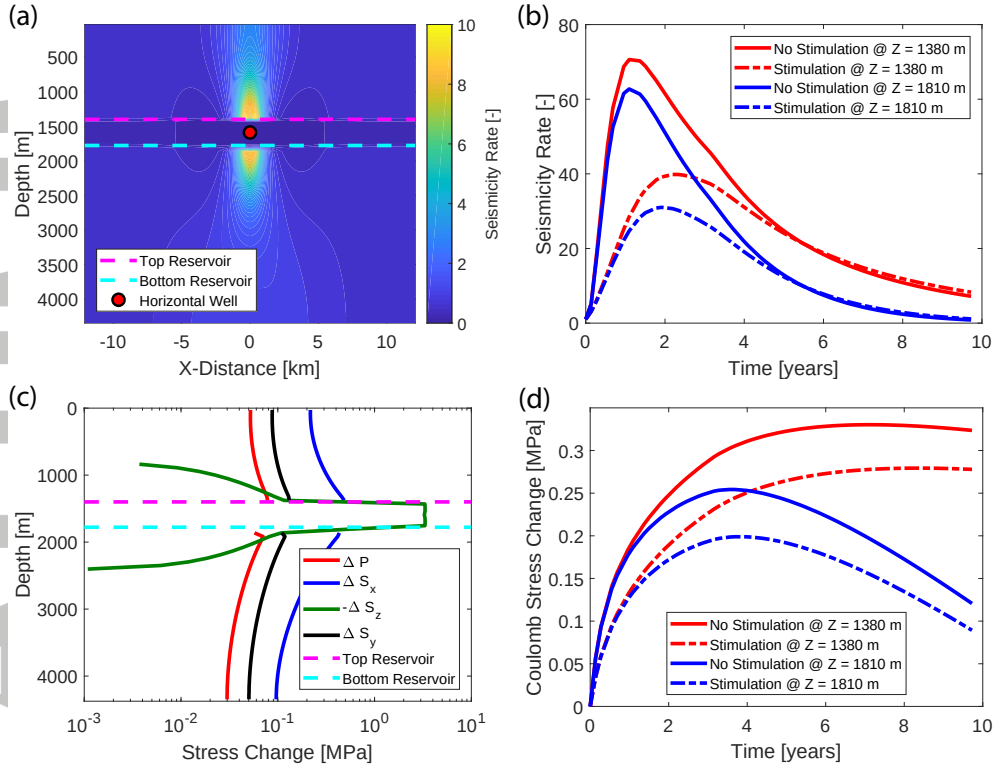


Figure 4. Results at the end of simulation ($t = 10$ years); (a) The seismicity rate occurring for a reverse faulting environment which has not undergone stimulation. Seismicity is concentrated above and below the producing reservoir as predicted by Segall [1989]. (b) A comparison between a stimulated and non-stimulated reservoir's seismicity rate. Two locations are shown, directly above the reservoir (Depth = 1380 m) and directly below the reservoir (Depth = 1810 m), in-line with the producing well. Note that based on (a) these are the most seismically active areas in the simulation. (c) The changes in stress and pore pressure that occur due to production from a non-stimulated reservoir. Stresses and pressures are taken in-line with the well in the vertical direction. The negative change in S_z is plotted because tension is induced at all relevant depths in this direction. (d) The Coulomb stress changes in the stimulated and non-stimulated cases. The comparison is made vertically, in-line with the producing well.

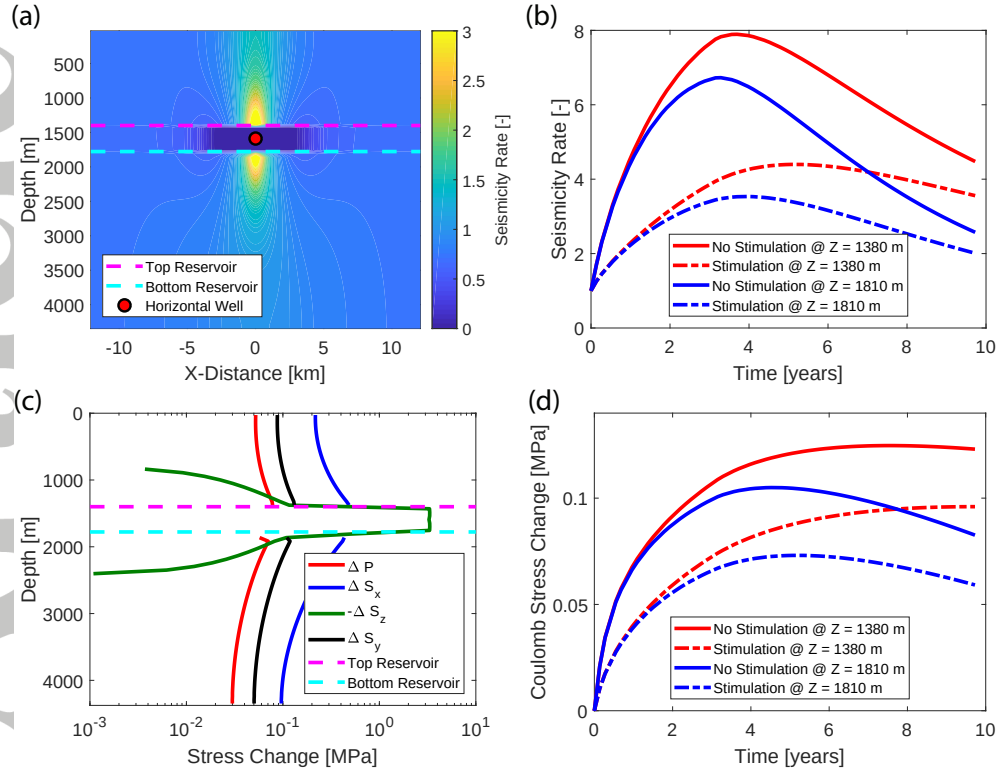


Figure 5. Results at the end of simulation ($t = 10$ years); (a) Seismicity rate occurring in a strike-slip faulting environment where the reservoir has not undergone stimulation. Seismicity is concentrated above and below the producing reservoir. (b) A comparison between a stimulated and non-stimulated reservoir's seismicity rate. Two locations are shown, directly above the reservoir (Depth = 1380 m) and directly below the reservoir (Depth = 1810 m), in-line with the producing well. Note that based on (a) these are the most seismically active areas in the simulation. (c) The changes in stress and pore pressure that occur due to production from a non-stimulated reservoir. Stresses and pressures are taken in-line with the well in the vertical direction. The negative change in S_z is plotted because tension is induced at all relevant depths in this direction. (d) The Coulomb stress changes between the stimulated and non-stimulated cases. The comparison is made vertically, in-line with the producing well.

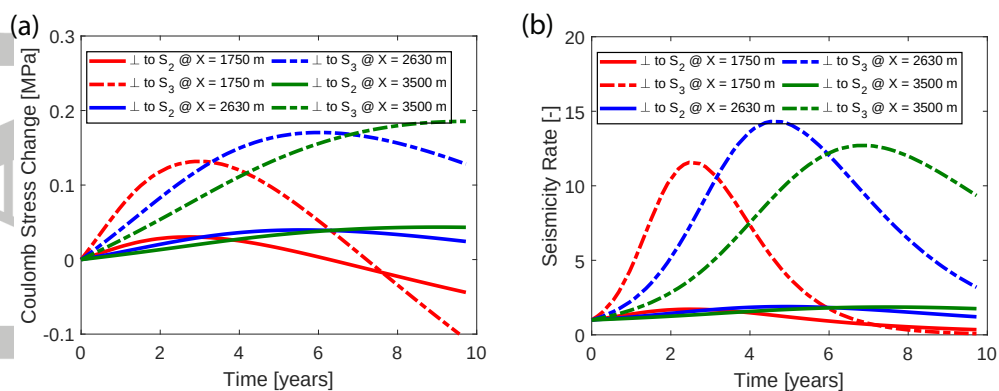


Figure 6. The effect of changing the direction that the horizontal well is drilled in in a normal faulting stress regime with no stimulation on the (a) Coulomb stress changes and (b) the seismicity rate. The legend indicates the orientation of the horizontal well with respect to the principal stresses. All values are taken at the same depth as the well at a specified horizontal distance from the well.

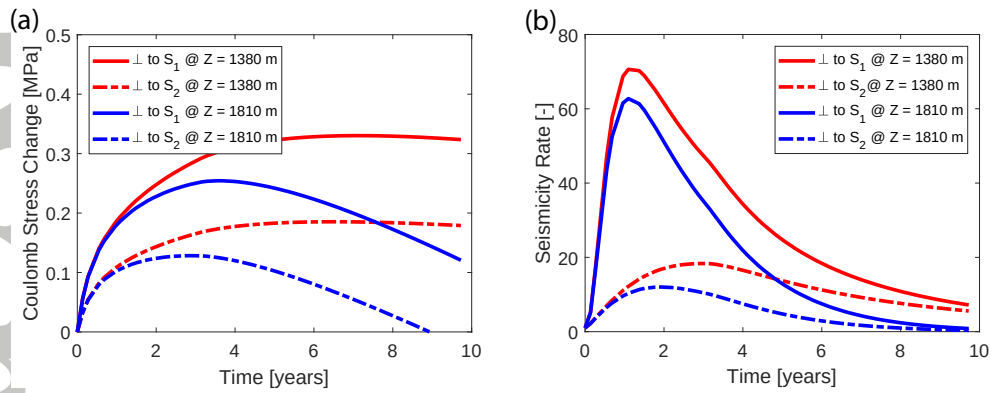


Figure 7. The effect of changing the direction that the horizontal well is drilled in in a reverse faulting stress regime with no stimulation on the (a) Coulomb stress changes and (b) the seismicity rate. For both images two locations are shown, directly above the reservoir (Depth = 1380 m) and directly below the reservoir (Depth = 1810 m), in-line with the producing well. The legend indicates the orientation of the horizontal well with respect to the principal stresses.

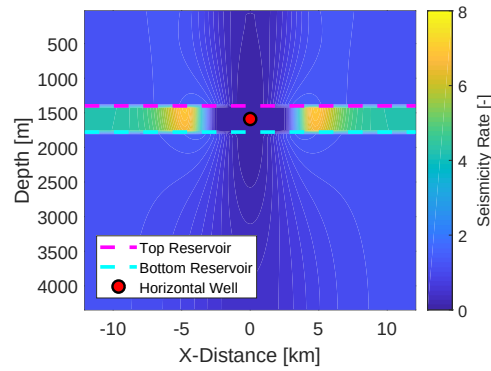


Figure 8. The seismicity occurring in a strike-slip faulting environment which has not undergone stimulation when the well is drilled parallel to the maximum principal stress. Note where the seismicity is located in this figure when compared to Figure 5a. The results shown are for the end of simulation ($t = 10$ years).

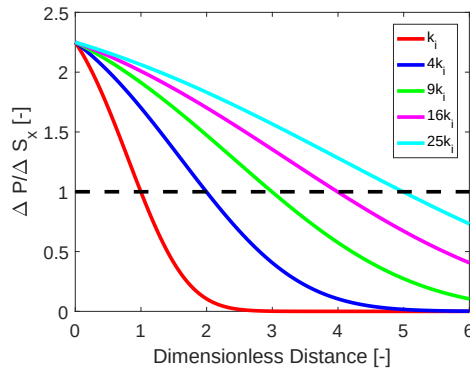


Figure 9. The ratio of changes in pore pressure to changes in horizontal stress for various permeabilities. The parameters are taken from the sandstone in Table 2, where k_i is equal to the sandstone permeability, and a dimensionless distance is defined as the distance required for the stress changes to be larger than pore pressure changes in the case where the permeability is equal to k_i . Note that the distance required for the ratio of pressure and stress changes to equal one increases with the square of permeability.